1. System Training and Knowledge Integration Approach

The proposed system employs a **Retrieval-Augmented Generation (RAG)** architecture to deliver evidence-based and personalized dietary recommendations. By integrating authoritative nutritional documents into a large language model (LLM), the system generates expert-level, contextually relevant responses tailored to individual patient profiles.

1.1 Retrieval-Augmented Generation Architecture

The RAG framework enhances the generative capabilities of the LLM by grounding outputs in verified nutritional literature. This ensures that recommendations align with established dietary standards while remaining flexible for patient-specific personalization. By combining retrieval-based knowledge with generative reasoning, the system produces accurate, clinically meaningful guidance.

1.2 Knowledge Base Construction

The knowledge base is constructed through a multi-stage document ingestion pipeline:

- **Document Ingestion:**
  Authoritative dietary guidelines, specifically the Dietary Reference Intakes (DRIs) issued by the National Academies of Sciences, Engineering, and Medicine (U.S.), are collected and preprocessed to form the system's nutritional knowledge base.

- **Text Extraction and Splitting:**
  Textual data is extracted and segmented into semantic chunks to improve retrieval accuracy.

- **Embedding Generation:**
  Each text chunk is transformed into a dense vector representation using the **nomic-embed-text** embedding model.

- **Vector Storage:**
  The resulting embeddings are stored in a **FAISS (Facebook AI Similarity Search) vector database**, forming a local knowledge base for semantic similarity search.

1.3 Patient-Specific Nutritional Profiling

Before invoking the LLM, the system calculates individualized nutritional targets, including **Body Mass Index (BMI)**, **Estimated Energy Requirement (EER)**, and **macronutrient distribution**. These deterministic outputs are embedded into the prompt context, ensuring that the LLM generates responses within patient-specific boundaries.

2. System Workflow

2.1 Knowledge Base Construction Workflow

This workflow illustrates the process of building the local knowledge base referenced by the LLM:

1. **Text Extraction:** Information is extracted from Dietary Reference Intakes (DRI) documents.
2. **Text Splitting:** Documents are divided into smaller semantic chunks.
3. **Embedding Generation:** Chunks are encoded into vectors using nomic-embed-text.
4. **Vector Storage:** Embeddings are stored in FAISS for retrieval.

2.2 Recommendation Generation Workflow

This workflow describes the real-time operation of the recommendation system:

1. **User Input:** Patient profile data—including height, weight, BMI, physical activity level, age, and sex—are collected and then used to compute nutritional requirements, such as the EER and macronutrient intake ranges.Meal intake information is recorded at the meal level, by consumed weight and volume, to support accurate dietary intake assessment.
2. **Query Vectorization:** The input is embedded to enable similarity-based retrieval.
3. **Retrieval:** A **Maximum Marginal Relevance (MMR)** search retrieves the most relevant nutritional knowledge from the FAISS store.
4. **Response Generation:** The **DeepSeek-R1:8b** model integrates retrieved knowledge with computed nutritional values to generate dietary recommendations.

3. Reference Literature

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